

The background of the slide is black, featuring several large, smooth, colorful sine waves in teal, orange, purple, and green that oscillate across the frame. A central black rectangular box contains the text.

TU Delft Aerospace Engineering

Waves and Vibrations

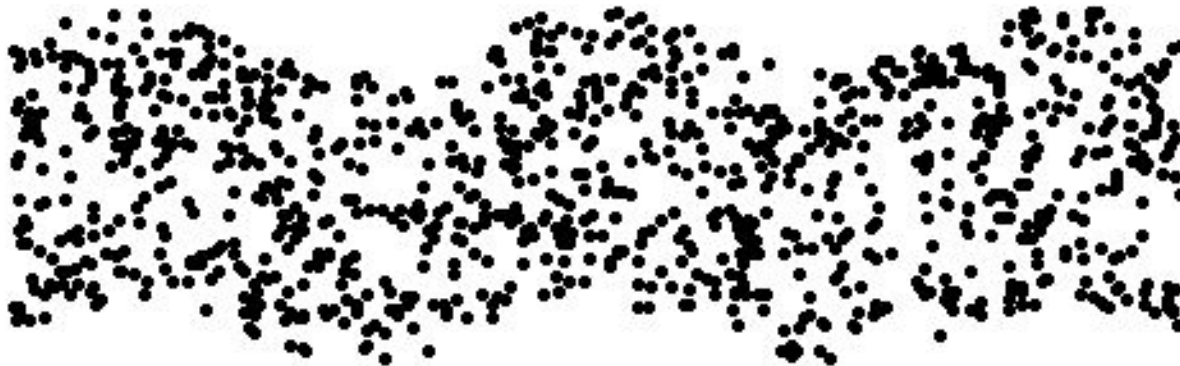
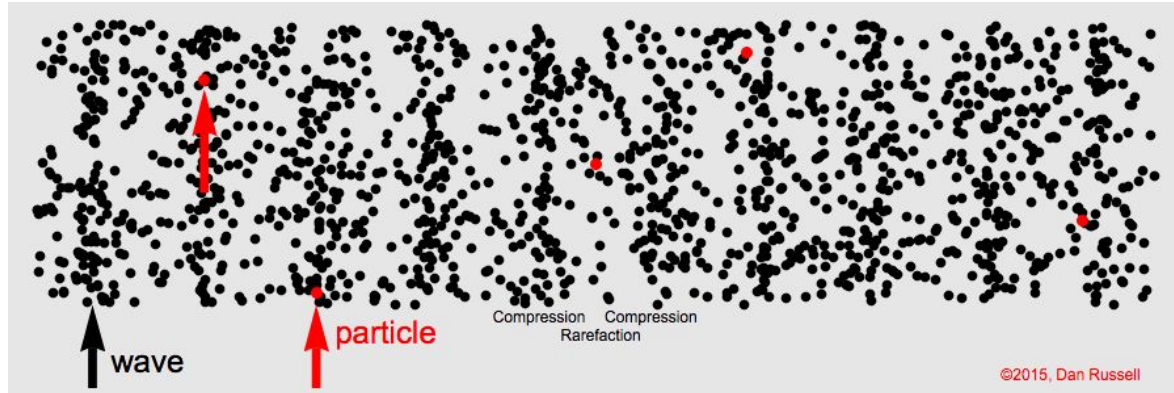
Marcos Carrera Valle

What is a Wave?

Waves transfer energy and information without transferring matter.

Types of Waves

- Longitudinal waves
 - Particles vibrate parallel to wave direction.
 - Example: Sound waves.
- Transverse waves
 - Particles vibrate perpendicular to wave direction.
 - Example: Light waves.



Key Wave Terms

Amplitude, Wavelength, Frequency, Period, Wavefront

Units

1. Amplitude
2. Wavelength
3. Frequency
4. Period

- A. λ
- B. f
- C. A
- D. T

Units

1. Amplitude
2. Wavelength
3. Frequency
4. Period

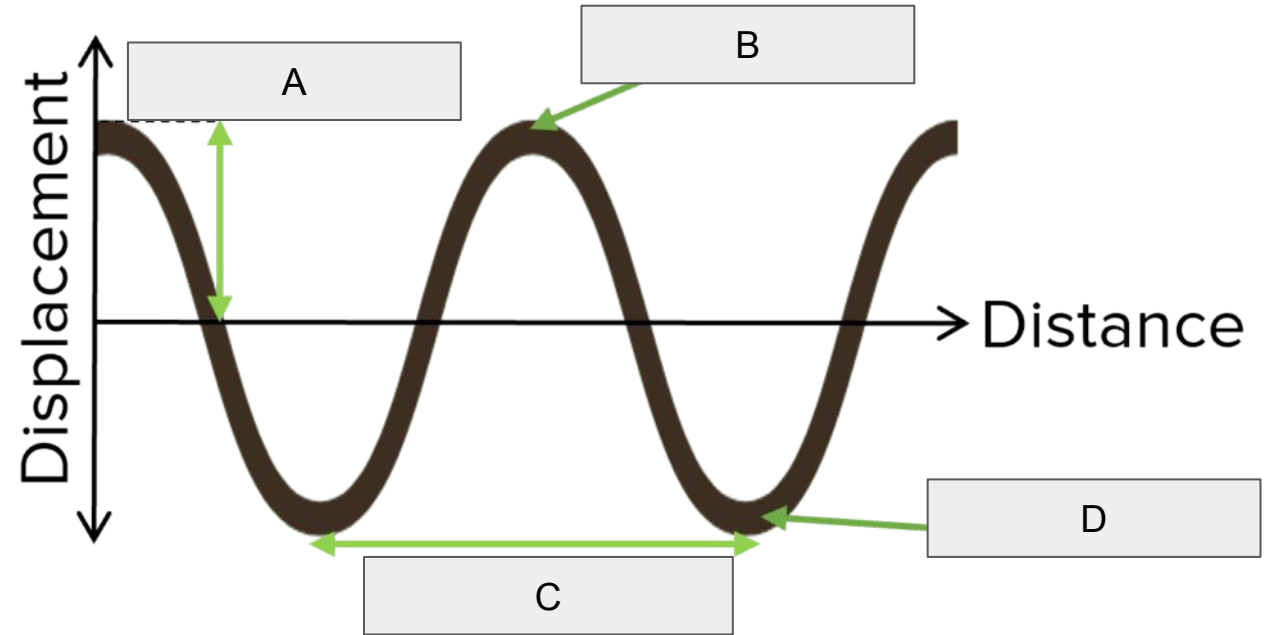
- A. Hz
- B. sec
- C. 1/sec
- D. m

Units

1. Amplitude, A [m]
 2. Wavelength, λ [m]
 3. Frequency, f [Hz]/ [1/sec]
 4. Period, T [sec]
-
- A. Distance between two identical points on a wave.
 - B. Time taken for one wave to pass a point.
 - C. Number of waves passing a point per second.
 - D. Maximum displacement from rest position.

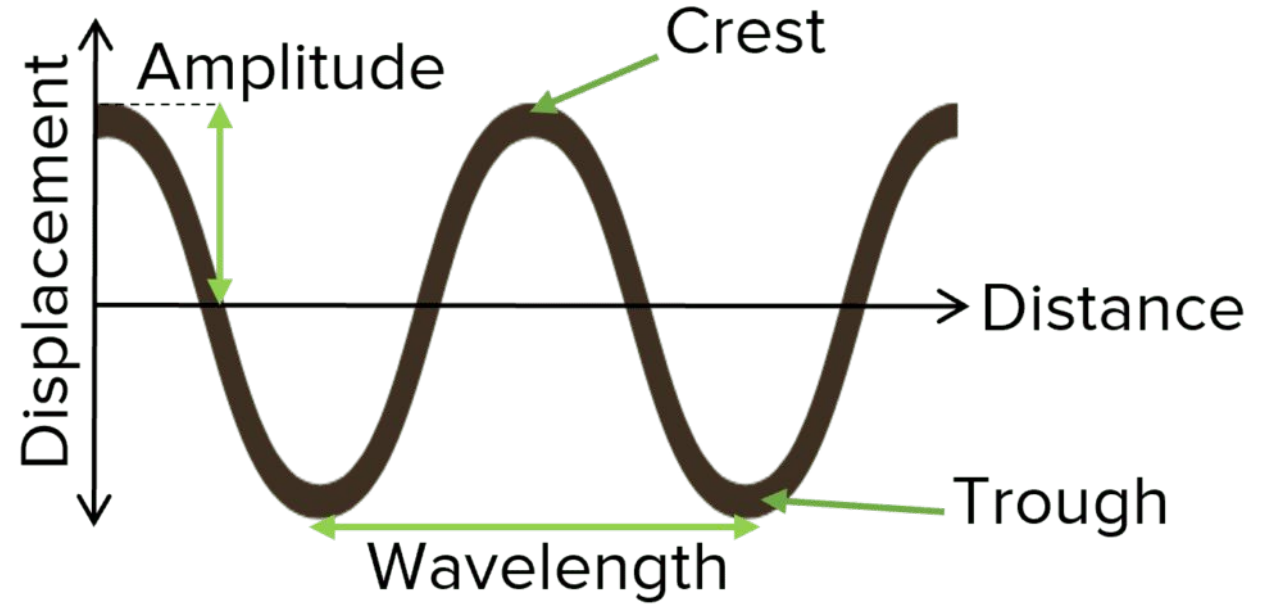
Units

1. Trough
2. Amplitude
3. Wavelength
4. Crest



Units

1. Trough
2. Amplitude
3. Wavelength
4. Crest



Formulae

Period: $T = 1 / f$

Wave Speed: $v = f \times \lambda$

Wave Speed equation

What is the velocity of a sound wave?

Wave Speed equation

What is the velocity of a sound wave? - **343 m/s**

Do you know what frequency of sound waves a human can hear?

Wave Speed equation

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Do you know what frequency of sound waves a human can hear? **20-20,000 Hz**

What is the range of wavelengths of sound a human can hear?

Wave Speed equation

What is the velocity of a sound wave? - **343 m/s**

Do you know what frequency of sound waves a human can hear? **20-20,000 Hz**

What is the range of wavelengths of sound a human can hear? **0.01715 - 17.15 m**

Waves

Consider four sinusoidal waves traveling through the same medium. The general equation for a displacement wave is $y(t) = A \sin(\omega t + \phi)$ where $\omega = 2\pi f$

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1) You are given the following two waves:

$$y_1(t) = 5 \sin(40\pi t)$$

$$y_2(t) = 3 \sin(80\pi t)$$

What is the frequency and period of each wave?

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$$y_2(t) = 3 \sin(80\pi t)$$

What is the frequency and period of each wave?

Since $\omega_1 = 40\pi$ and $\omega_1 = 2\pi f$, then $f_1 = 20$ Hz. Therefore, $T_1 = 0.05$ s.

Since $\omega_2 = 80\pi$ and $\omega_2 = 2\pi f$, then $f_2 = 40$ Hz. Therefore, $T_1 = 0.0025$ s.

Waves

Consider four sinusoidal waves traveling through the same medium. The general equation for a displacement wave is $y(t) = A \sin(\omega t + \phi)$ where $\omega = 2\pi f$

2) A third wave is introduced:

$$y_3(t) = 3 \sin(80\pi t + \pi/2)$$

Can this wave constructively interfere with wave 1?

What is the phase difference between y_2 and y_3 ?

Briefly describe the resulting motion if Wave 2 and Wave 3 are superposed.

Waves

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$$y_3(t) = 3 \sin(80\pi t + \pi/2)$$

Can this wave constructively interfere with wave 1? **No, different frequencies**

What is the phase difference between y_2 and y_3 ? **$\pi/2$**

Briefly describe the resulting motion if Wave 2 and Wave 3 are superposed.

The combined wave will still oscillate at the same frequency

Because the waves are not perfectly in phase (constructive) nor perfectly out of phase (destructive), they interfere to create a new amplitude $A_4 = (A_2^2 + A_3^2)^{1/2} = 4.24$. The new wave will have a phase shift exactly halfway between the two originals

Waves

Consider four sinusoidal waves traveling through the same medium. The general equation for a displacement wave is $y(t) = A \sin(\omega t + \phi)$ where $\omega = 2\pi f$

3) A fourth wave, which shares the same frequency as Wave 1, is given:

$$y_4(t) = 2 \sin(40\pi t)$$

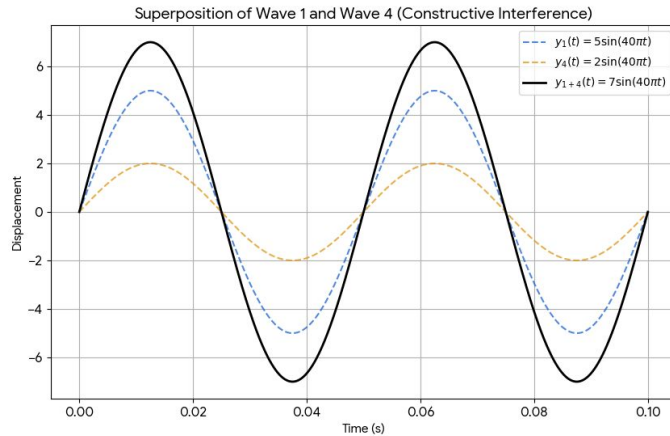
Show that the superposition of Wave 1 and Wave 4 results in constructive interference (Make a plot)

Write the equation for the resulting combined wave, $y_{1+4}(t)$

Derive the equation for a Wave 5 that, when added to the combined y_{1+4} , will result in total destructive interference

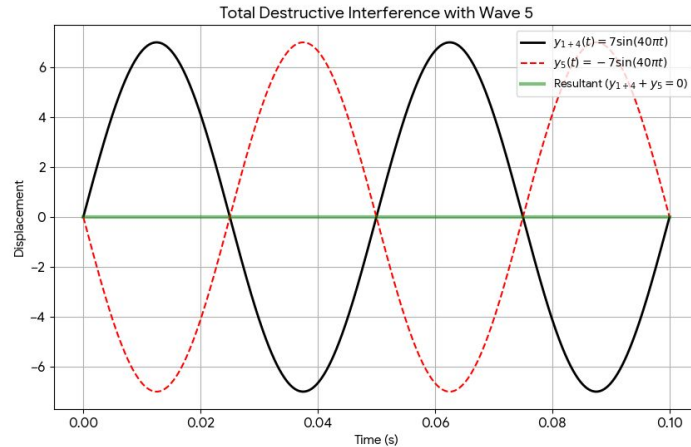
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Interference

Two identical speakers, S1 and S2, are placed 4.0 meters apart at $y = 0$ m. They are connected to the same signal generator and oscillate in phase, emitting a constant sound wave with a frequency of 680 Hz.

A student stands at Point P, which is directly in front of Speaker S1 and at $y = 5$ m

Does the student at Point P hear a loud sound (constructive interference) or a quiet sound (destructive interference)? Explain using the path difference.

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If he moves to the left slowly, will he hear the sound louder or quieter?

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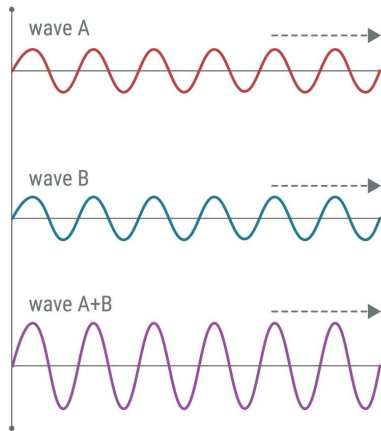
If he moves to the left slowly, will he hear the sound louder or quieter?

By how much does he have to move to the left to reach the point of loudest noise?

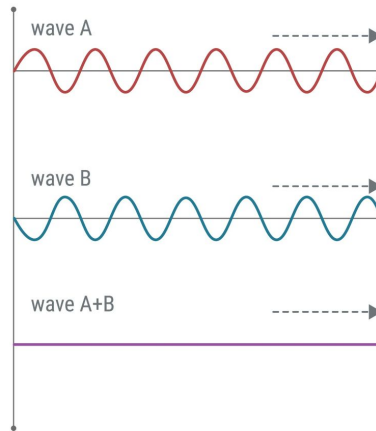
Wave Interference

Wave Interference

constructive interference



destructive interference

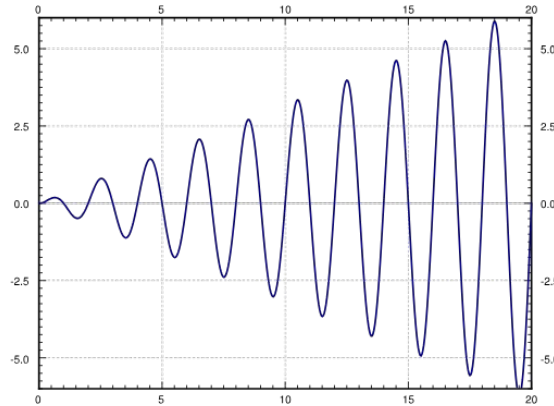


Resonance

Resonance occurs when a system is driven by an external force at its natural frequency.

When the frequency of the external force matches this natural frequency, the system absorbs energy much more efficiently.

This leads to a dramatic increase in the amplitude of the oscillation.



Resonance

The natural frequency of a system is given by
 $\omega_n = \sqrt{k/m}$ (Mechanical) or
 $\sqrt{g/L}$ (Pendulum)

Resonance

A small wind turbine is modeled as a vertical steel beam with a heavy generator (nacelle) at the top.

Mass of the nacelle: 500 kg

Stiffness of the tower: 20,000 N/m

External Wind: A gusty wind is hitting the turbine. The wind speed fluctuates, creating a periodic force with a frequency of 0.95 Hz.

Calculate the natural frequency of the wind turbine system in Hertz
Determine if the turbine is at risk of resonance.

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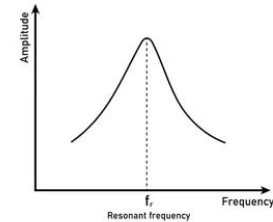
External Wind: A gusty wind is hitting the turbine. The wind speed fluctuates, creating a periodic force with a period of 1.05 s.

Calculate the natural frequency of the wind turbine system in Hertz
Determine if the turbine is at risk of resonance.

$$f_0 = 1.01 \text{ Hz}$$

High risk of resonance (within 10% of f_0)

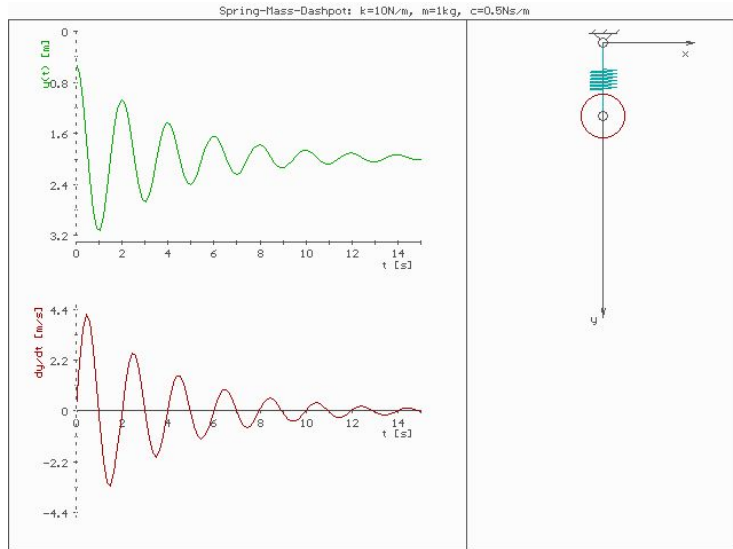
What design change would you implement?



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Damping

Damping is the process by which energy is removed from an oscillating system, usually through friction or air resistance.



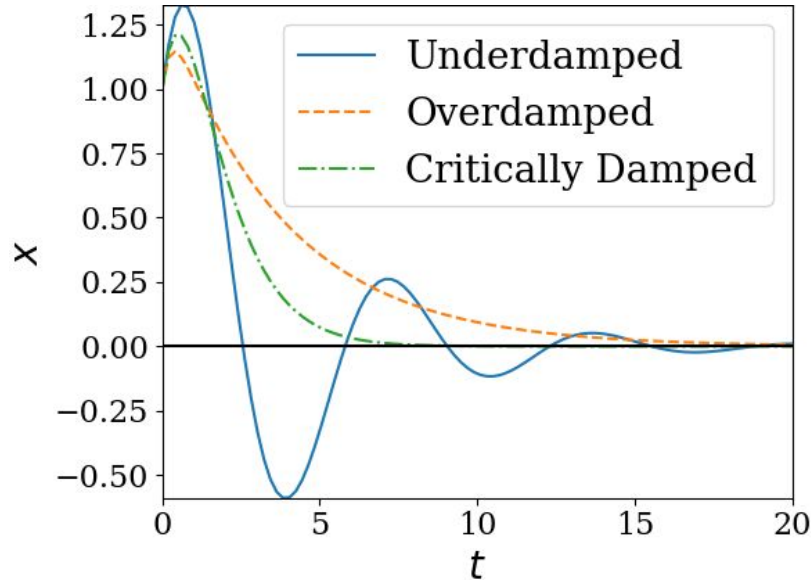
Damping

Engineers categorize damping based on how quickly the system returns to rest:

- 1) Underdamped: The system oscillates back and forth, with the size of the swings gradually getting smaller (e.g., a guitar string).
- 2) Critically Damped: The system returns to its equilibrium position as quickly as possible without any "overshoot" or extra swings. This is the ideal goal for car suspension.
- 3) Overdamped: The system is so heavily restricted that it takes a very long time to return to equilibrium (e.g., a door closer that moves extremely slowly).

Damping

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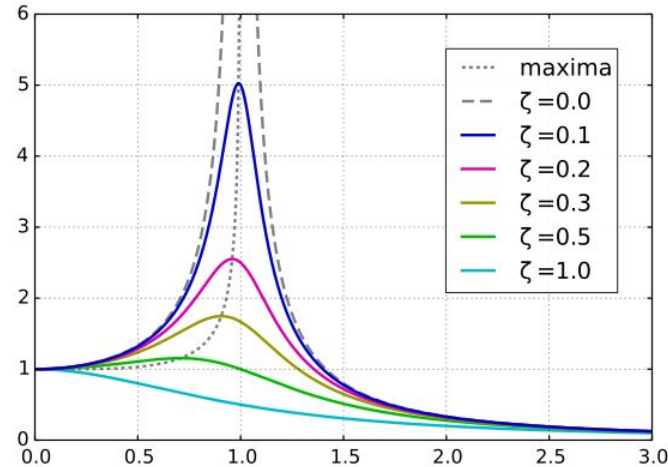
Damping

In physics and engineering, we use the damping ratio $\zeta = c/c_c$, where c is your damping coefficient and c_c is the critical damping coefficient $c_c = 2\sqrt{mk}$

If $\zeta < 1$: Underdamped

If $\zeta = 1$: Perfectly damped

If $\zeta > 1$: Overdamped



Damping

Going back to our wind turbine problem:
Actual Damping Coefficient: 1,200 Ns/m

Find the Critical Damping Coefficient

Based on c and c_c , is the turbine currently underdamped, overdamped, or critically damped?

To avoid the 0.95 Hz wind resonance, we need to change the natural frequency.
Is it more effective to change the Stiffness or the Mass?

Damping

Going back to our wind turbine problem:
Actual Damping Coefficient: 1,200 Ns/m

Find the Critical Damping Coefficient: **6,324.56**

Based on c and c_c , is the turbine currently underdamped, overdamped, or critically damped? **Underdamped**

To avoid the 0.95 Hz wind resonance, we need to change the natural frequency.
Is it more effective to change the Stiffness or the Mass?

Changing Stiffness is usually the preferred and most effective solution.

Simple Harmonic Motion

Simple Harmonic Motion (SHM) is a specific type of periodic motion where an object oscillates back and forth around a central equilibrium position.

For a system to be in simple harmonic motion, it must satisfy these conditions:

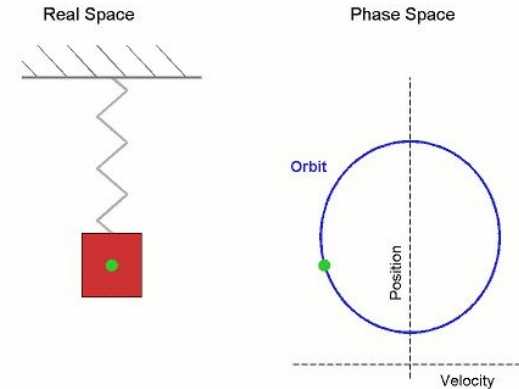
- 1) Equilibrium Position: There is a "rest" position where the net force is zero.
- 2) Restoring Force: If moved, a force tries to pull it back (like a spring or gravity).
- 3) Proportionality: The further you pull it, the harder the force pulls back

To describe SHM, we use amplitude, period, frequency and phase

Simple Harmonic Motion

One of the most beautiful parts of physics is that Simple Harmonic Motion is just a 1D projection of circular motion.

This is why the equation for harmonic motion always uses sine or cosine such as $x(t) = A \cos(\omega t + \phi)$



Simple Harmonic Motion

Real life examples of SHM:

- A Pendulum: A clock's weight swinging back and forth.
- A Mass on a Spring: Bouncing up and down.
- Guitar Strings: When plucked, the string vibrates in harmonic motion to create sound waves.
- Atoms in a Solid: Atoms actually vibrate around their "fixed" positions in a crystal lattice similarly to little masses on springs.

Whiteboard

Simple Harmonic Motion

A sensor with a mass of $m=2$ kg is mounted on a bracket that acts like a spring with a stiffness of $k = 800$ N/m. Initially we assume there is no friction. The sensor is pulled down 0.1 m and released.

Calculate the natural angular frequency ω_n and the cyclic frequency f_n .
Write the equation for the displacement $x(t)$.
Sketch the displacement curve for the first two periods.

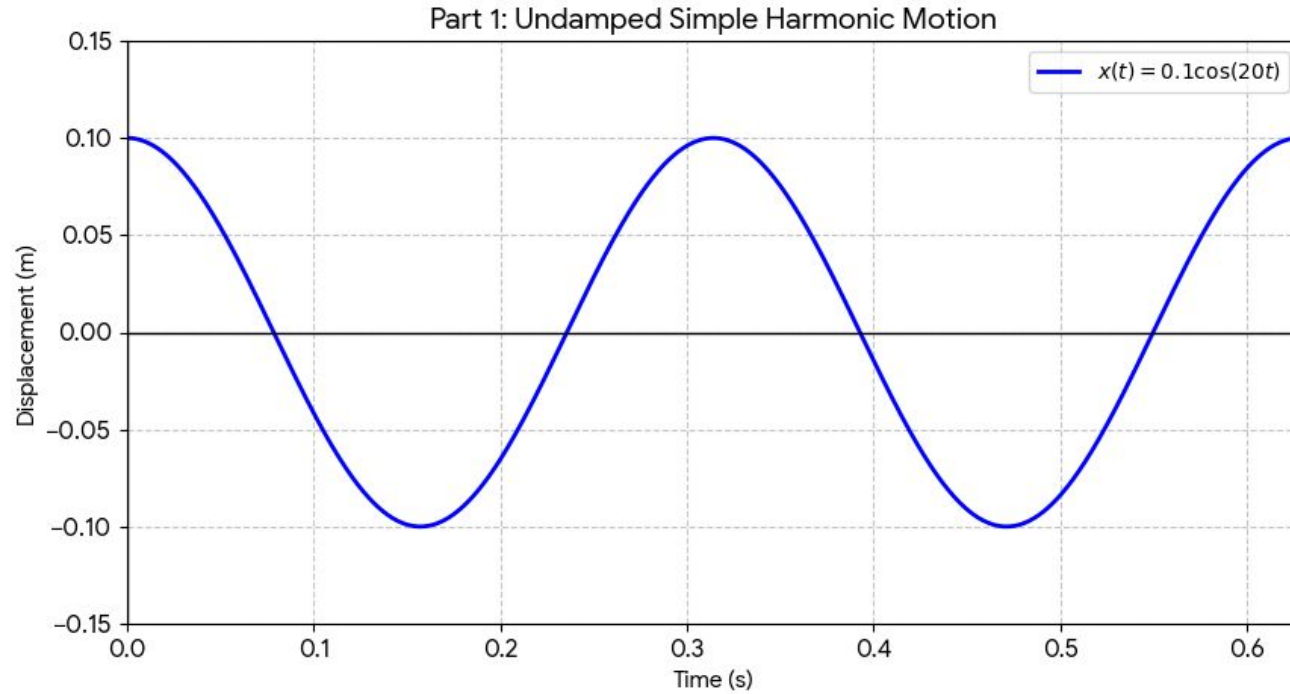
Simple Harmonic Motion

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Calculate the natural angular frequency ω_n and the cyclic frequency f_n .
Write the equation for the displacement $x(t)$.
Sketch the displacement curve for the first two periods.

$$\omega_n = 20 \text{ rad/s}, f_n = 3.18 \text{ Hz}, x(t) = 0.1\cos(20t)$$

Simple Harmonic Motion



Simple Harmonic Motion

A sensor with a mass of $m=2$ kg is mounted on a bracket that acts like a spring with a stiffness of $k = 800$ N/m. Initially we assume there is no friction. The sensor is pulled down 0.1 m and released.

In reality, the mount is housed in a protective oil casing that provides a damping coefficient of 20 Ns/m

Calculate the critical damping coefficient c_c .

Find the damping ratio ζ and identify the type of damping

How does the displacement curve change compared to the previous SHM?

Simple Harmonic Motion

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Calculate the critical damping coefficient c_c .

Find the damping ratio ζ and identify the type of damping

How does the displacement curve change compared to the previous SHM?

$c_c = 80$ N/sm, $\zeta = 0.25$ (Underdamped), the sensor will still oscillate, but the amplitude will decay exponentially over time.

Homework

Homework

- 1) Identify whether the following waves are Longitudinal or Transverse:
- A. Light waves traveling through a vacuum.
 - B. Sound waves traveling through the air.
 - C. A wave pulse sent down a slinky by moving it back and forth (parallel to the spring).
 - D. A "human wave" in a sports stadium where fans stand up and sit down.
 - E. Ripples on the surface of a pond.

Homework

2)

- a) If the frequency of a sound wave increases, what happens to the pitch that we hear?
- b) If the amplitude of a sound wave increases, what happens to the volume (loudness)?
- c) Does sound travel faster through a solid (like a steel bar) or a gas (like air)? Why?

Homework

3) Evaluate the following statements regarding wave behavior:

Statement 1: Mechanical waves (like sound) can travel through the vacuum of outer space. (True / False)

Statement 2: In a longitudinal wave, the particles of the medium move perpendicular to the direction of the wave's travel. (True / False)

Statement 3: Period is the amount of time it takes for one complete cycle of a vibration to occur. (True / False)

Homework

4) A wave on a string has a frequency of 50 Hz and a wavelength of 0.4 meters. Calculate the speed of the wave.

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Is this an electromagnetic wave?

Homework

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Is this an electromagnetic wave?

Radio waves can also have a wavelength of around 1 meter. What is the period of a radio wave?

Homework

5) A transverse wave traveling along a wire is described by the following displacement function (where distances are in meters and time is in seconds):

$$y = 0.05 \sin(40\pi t - 2\pi x)$$

Based on this function, determine the following:

The Amplitude: What is the maximum displacement of the wave?

The Frequency: Using the time component (40π), calculate the frequency in Hz.

The Period: Calculate the time it takes for one full cycle to pass.

The Wave Speed: Calculate the velocity at which the wave is traveling.

Homework

6) Two identical wave sources, Wave A and Wave B, are sent toward each other. Both have a frequency of 100 Hz and an amplitude of 2.0 cm.

Scenario 1: If the waves meet exactly "in phase" (crest meets crest), what is the resulting amplitude and the new period of the combined wave?

Scenario 2: If Wave B is delayed by exactly 0.005 seconds, how will this affect the interference when they meet?

Homework

7)

Two coherent point sources, S1 and S2, are separated by a distance "d" in a vacuum. They emit monochromatic light with a wavelength of 633 nm. An interference pattern is observed on a screen placed parallel to the plane of the sources at a distance $D = 2.50$ m.

The distance between the central maximum and the third-order maximum ($n = 3$) on the screen is measured to be 3.80 cm.

- a) Calculate the separation "d" between the two sources in millimeters (mm).
- b) Determine the path difference for the light reaching the second-order minimum

Homework

8)

During the launch of a sounding rocket, the liquid oxygen in the cylindrical tank oscillates. Engineers model the movement of the liquid's center of mass as a Simple Pendulum. The "effective length" of this liquid pendulum is determined to be $L = 0.45$ m. The rocket's engine vibrates at a frequency of 22 Hz, but it also produces a lower-frequency "pogo" oscillation at 0.74 Hz.

- a) Calculate the natural frequency of the liquid sloshing using the simple pendulum model.
- b) Determine which engine vibration (22 Hz or 0.74 Hz) poses a structural risk due to resonance.
- c) If the rocket accelerates at 3g during ascent, calculate the new natural frequency of the sloshing.

Homework

9)

An aerospace sensor is modeled as a mass-spring system undergoing damped oscillations. If the amplitude of the oscillation decreases by 50% every 4 full cycles, which of the following best describes the energy of the system after those 4 cycles?

Homework

10)

A block of mass $M = 4.0 \text{ kg}$ is attached to a horizontal spring with a spring constant $k = 400 \text{ N/m}$. The block slides on a frictionless surface. The system is pulled 0.2 m from its equilibrium position and released from rest at $t = 0$.

At the exact moment the block passes through the equilibrium position, a second mass $m = 2.0 \text{ kg}$ is dropped vertically onto the moving block and sticks to it (a perfectly inelastic collision).

- a) Calculate the period of the oscillation before the second mass is added.
- b) Calculate the new amplitude of the system after the collision.
- c) Calculate the percentage change in the total mechanical energy of the system caused by the collision.

Homework

1) Identify whether the following waves are Longitudinal or Transverse:

A. Light waves traveling through a vacuum. **Transverse**

B. Sound waves traveling through the air. **Longitudinal**

C. A wave pulse sent down a slinky by moving it back and forth (parallel to the spring). **Longitudinal**

D. A "human wave" in a sports stadium where fans stand up and sit down.
Transverse

E. Ripples on the surface of a pond. **Longitudinal**

Homework

2)

a) If the frequency of a sound wave increases, what happens to the pitch that we hear? **Increases**

b) If the amplitude of a sound wave increases, what happens to the volume (loudness)? **Louder**

c) Does sound travel faster through a solid (like a steel bar) or a gas (like air)? Why? **Sound travels faster in solids because particles are more tightly packed, allowing the vibration to transfer more quickly.**

Homework

3) Evaluate the following statements regarding wave behavior:

Statement 1: Mechanical waves (like sound) can travel through the vacuum of outer space. (True / **False**)

Statement 2: In a longitudinal wave, the particles of the medium move perpendicular to the direction of the wave's travel. (True / **False**)

Statement 3: Period is the amount of time it takes for one complete cycle of a vibration to occur. (**True** / False)

Homework

4) A wave on a string has a frequency of 50 Hz and a wavelength of 0.4 meters. Calculate the speed of the wave.

20 m/s

Homework

4) A wave on a string has a frequency of 50 Hz and a wavelength of 0.4 meters. Calculate the speed of the wave. **20 m/s**

Is this an electromagnetic wave? **No, EM waves travel at $3 \cdot 10^8$ m/s**

Radio waves can also have a wavelength of around 1 meter. What is the period of a radio wave with wavelength 0.4 meters?

$1.33 \cdot 10^{-9}$

Homework

5) A transverse wave traveling along a wire is described by the following displacement function (where distances are in meters and time is in seconds):

$$y = 0.05 \sin(40\pi t - 2\pi x)$$

Based on this function, determine the following:

The Amplitude: What is the maximum displacement of the wave? **0.05 m**

The Frequency: Using the time component (40π), calculate the frequency in Hz. **20 Hz**

The Period: Calculate the time it takes for one full cycle to pass. **0.05 s**

The Wave Speed: Calculate the velocity at which the wave is traveling. **20 m/s**

Homework

6) Two identical wave sources, Wave A and Wave B, are sent toward each other. Both have a frequency of 100 Hz and an amplitude of 2.0 cm.

Scenario 1: If the waves meet exactly "in phase" (crest meets crest), what is the resulting amplitude and the new period of the combined wave? **A = 4 cm, T = 0.01s**

Scenario 2: If Wave B is delayed by exactly 0.005 seconds, how will this affect the interference when they meet? **T = 0.01, Delay = Half a period - Total Destructive Interference**

Homework

7)

a) Calculate the separation "d" between the two sources in millimeters (mm).

Find fringe spacing (y). Since the distance to the 3rd maximum is 3.80 cm, one fringe (y) = $3.80 / 3 = 1.267$ cm.

Convert to meters: $y = 0.01267$ m and use the formula $d = (\lambda * D) / y$.

$d = (633 * 10^{-9} * 2.50) / 0.01267 = 0.125$ mm.

b) Determine the path difference for the light reaching the second-order minimum ($n = 1.5$).

For destructive interference (minima), path difference = $(n + 0.5) * \text{wavelength}$.

For the second-order minimum, we use $(1 + 0.5) = 1.5$.

Path Difference = $1.5 * 633 \text{ nm} = 949.5 \text{ nm}$

Homework

8)

a) Calculate the natural frequency of the liquid sloshing using the simple pendulum model.

$$f_n = \sqrt{g/L} = 0.743 \text{ Hz}$$

b) Determine which engine vibration (22 Hz or 0.74 Hz) poses a structural risk due to resonance.

0.74 Hz poses a risk as it is very close to the natural frequency

c) If the rocket accelerates at 3g during ascent, calculate the new natural frequency of the sloshing.

$$f_n = \sqrt{3g/L} = 1.29 \text{ Hz}$$

Homework

9)

An aerospace sensor is modeled as a mass-spring system undergoing damped oscillations. If the amplitude of the oscillation decreases by 50% every 4 full cycles, which of the following best describes the energy of the system after those 4 cycles?

The total energy is 25% of its initial value, as energy is proportional to the square of the amplitude

Homework

10)

a) Calculate the period of the oscillation before the second mass is added.

$$T = 2\pi\sqrt{M/k} = 0.628 \text{ s}$$

b) Calculate the new amplitude of the system after the collision.

$$V_{\max} = A\sqrt{k/M} = 2 \text{ m/s}, V_{\text{new}} = 1.33 \text{ m/s}, A_{\text{new}} = 0.163 \text{ m}$$

c) Calculate the percentage change in the total mechanical energy of the system caused by the collision.

Loss of 33.4% of total energy